

HandsOn-1

Objectives

Familiar with Git commands like git init, git status, git add, git commit, git push, and git pull.

In this hands-on lab, you will learn how to

- Setup your machine with Git Configuration
- Integrate notepad++.exe to Git and make it a default editor
- Add a file to source code repository

Commands to add in created Repo

git --version

- ◆ Checks if Git is installed and shows the installed version.

git config --global user.name "Lavanya-Yannam"

- ◆ Sets your Git username globally for all repositories on your system.

git config --global user.email "your_personal_email@example.com"

- ◆ Sets your Git email globally for all repositories on your system.

git config --global --list

- ◆ Shows all global Git configuration settings (name, email, editor, etc.).

mkdir GitDemo

- ◆ Creates a new folder named GitDemo.

cd GitDemo

- ◆ Changes the current directory to GitDemo.

git init

- ◆ Initializes a new empty Git repository in the current folder.

ls -a

- ◆ Lists all files, including hidden ones like .git/.

echo "welcome to Git Training" > welcome.txt

- ◆ Creates a file welcome.txt and writes the text into it.

ls

- ◆ Lists visible files in the current directory.

cat welcome.txt

- ◆ Displays the content of the file welcome.txt.

git status

- ◆ Shows the current state of the working directory and staging area.

git add welcome.txt

- ◆ Stages the file welcome.txt for commit.

git commit -m "Added welcome.txt file - contains welcome message for Git training"

- ◆ Commits the staged file with a single-line commit message.

git commit

- ◆ Opens the default editor for writing a multi-line commit message.

→ Authentication & pushing (two recommended options)

HTTPS with Personal Access Token (PAT)

1. Create a PAT on GitHub (Settings → Developer settings → Personal access tokens → Generate new token) — give repo scope. **Copy the token.**
2. Remove any cached wrong credentials:
 - In Windows: **Control Panel → Credential Manager → Windows Credentials** → remove entries for github.com.
 - Or in Git Bash: git credential-cache exit

git remote set-url origin https://github.com/Lavanya-Yannam/GitDemo.git

- ◆ Updates the origin remote to point to your GitHub repository URL.

git remote -v

- ◆ Displays the current remote repository URLs for fetch and push operations.

git push origin master

- ◆ Uploads your local commits from the master branch to the origin remote repository.

❓ When asked for username: enter Lavanya-Yannam

❓ When asked for password: **paste the PAT** (not your GitHub password).

After a successful push, open the git repo in the github, You should see welcome.txt listed.

Output screens:

```
MINGW64/c/Users/LAVANYA

LAVANYA@LAPTOP-7H755M1L MINGW64 ~
$ git config --global --list
user.name=Lavanya-Yannam
user.email=yannamlavanya2005@gmail.com
user.mail=lavanyayannamlavanya@gmail.com
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true

LAVANYA@LAPTOP-7H755M1L MINGW64 ~
$ notepad++
alias np='notepad++'
notepad++ ~/.bashrc
alias np='notepad++'

LAVANYA@LAPTOP-7H755M1L MINGW64 ~
$ alias np='notepad++'

LAVANYA@LAPTOP-7H755M1L MINGW64 ~
$ notepad++ ~/.bashrc

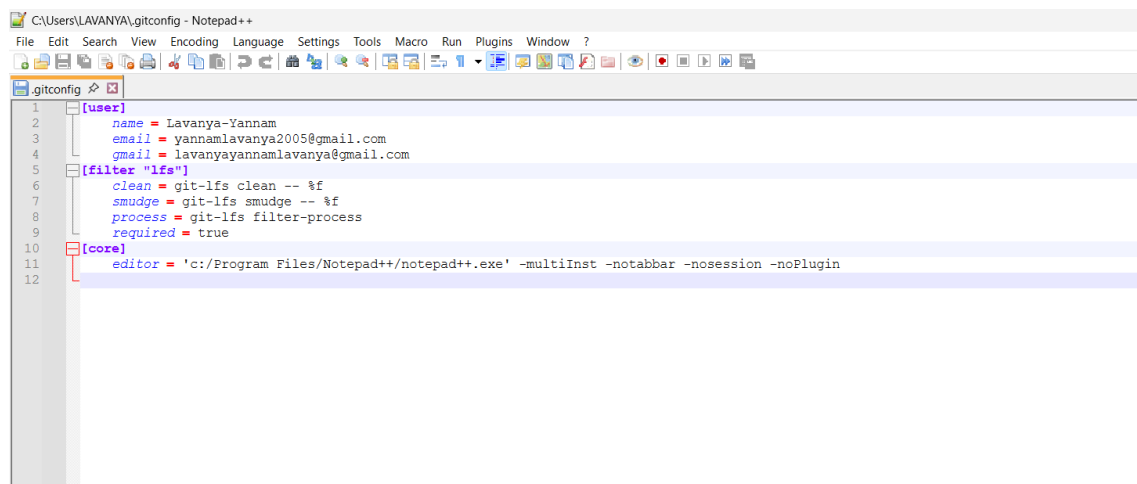
LAVANYA@LAPTOP-7H755M1L MINGW64 ~
$ alias np='notepad++'

LAVANYA@LAPTOP-7H755M1L MINGW64 ~
$

LAVANYA@LAPTOP-7H755M1L MINGW64 ~
$ source ~/.bashrc

LAVANYA@LAPTOP-7H755M1L MINGW64 ~
$ git config --global core.editor "'c:/Program Files/Notepad++/notepad++.exe' -multiInst -notabbar
r -nosession -noPlugin"

LAVANYA@LAPTOP-7H755M1L MINGW64 ~
$ git config --global -e
hint: waiting for your editor to close the file... |
```



The screenshot shows the Notepad++ application window with the file path 'C:\Users\LAVANYA\.gitconfig - Notepad++'. The file content is as follows:

```
1 [user]
2   name = Lavanya-Yannam
3   email = yannamlavanya2005@gmail.com
4   mail = lavanyayannamlavanya@gmail.com
5 [filter "lfs"]
6   clean = git-lfs clean -- %f
7   smudge = git-lfs smudge -- %f
8   process = git-lfs filter-process
9   required = true
10 [core]
11   editor = 'c:/Program Files/Notepad++/notepad++.exe' -multiInst -notabbar -nosession -noPlugin
12
```

```
MINGW64/c/Users/LAVANYA/GitDemo

LAVANYA@LAPTOP-7H755M1L MINGW64 ~
$ mkdir GitDemo

LAVANYA@LAPTOP-7H755M1L MINGW64 ~
$ cd GitDemo

LAVANYA@LAPTOP-7H755M1L MINGW64 ~/GitDemo
$ git init
Initialized empty Git repository in C:/Users/LAVANYA/GitDemo/.git/

LAVANYA@LAPTOP-7H755M1L MINGW64 ~/GitDemo (master)
$ ls -a
./ ../ .git/

LAVANYA@LAPTOP-7H755M1L MINGW64 ~/GitDemo (master)
$ echo "welcome to Git Training" > welcome.txt

LAVANYA@LAPTOP-7H755M1L MINGW64 ~/GitDemo (master)
$ ls
welcome.txt

LAVANYA@LAPTOP-7H755M1L MINGW64 ~/GitDemo (master)
$ cat welcome.txt
welcome to Git Training

LAVANYA@LAPTOP-7H755M1L MINGW64 ~/GitDemo (master)
$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
    welcome.txt

nothing added to commit but untracked files present (use "git add" to track)

LAVANYA@LAPTOP-7H755M1L MINGW64 ~/GitDemo (master)
$ git add welcome.txt
warning: in the working copy of 'welcome.txt', LF will be replaced by CRLF the next time Git touches it

LAVANYA@LAPTOP-7H755M1L MINGW64 ~/GitDemo (master)
$ git commit
hint: Waiting for your editor to close the file...
```

```
C:\Users\LAVANYA\GitDemo\.git\COMMIT_EDITMSG - Notepad++
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?

COMMIT_EDITMSG ✕
1
2 # Please enter the commit message for your changes. Lines starting
3 # with '#' will be ignored, and an empty message aborts the commit.
4 #
5 # On branch master
6 #
7 # Initial commit
8 #
9 # Changes to be committed:
10 #   new file:   welcome.txt
11 #
12 Added welcome.txt file
13 Contains welcome message for Git training
14
```

```

LAVANYA@LAPTOP-7H755M1L MINGW64 ~/GitDemo (master)
$ git status
On branch master
nothing to commit, working tree clean

LAVANYA@LAPTOP-7H755M1L MINGW64 ~/GitDemo (master)
$ git remote set-url origin https://github.com/Lavanya-Yannam/GitDemo.git

LAVANYA@LAPTOP-7H755M1L MINGW64 ~/GitDemo (master)
$ git remote -v
origin https://github.com/Lavanya-Yannam/GitDemo.git (fetch)
origin https://github.com/Lavanya-Yannam/GitDemo.git (push)

LAVANYA@LAPTOP-7H755M1L MINGW64 ~/GitDemo (master)
$ git push origin master
remote: Permission to Lavanya-Yannam/GitDemo.git denied to LavanyaYannam.
fatal: unable to access 'https://github.com/Lavanya-Yannam/GitDemo.git/': The requested URL retur
ned error: 403

LAVANYA@LAPTOP-7H755M1L MINGW64 ~/GitDemo (master)
$ git config --global user.name
Lavanya-Yannam

LAVANYA@LAPTOP-7H755M1L MINGW64 ~/GitDemo (master)
$ ^C

LAVANYA@LAPTOP-7H755M1L MINGW64 ~/GitDemo (master)
$ git credential-cache exit

LAVANYA@LAPTOP-7H755M1L MINGW64 ~/GitDemo (master)
$ git remote set-url origin https://github.com/Lavanya-Yannam/GitDemo.git

LAVANYA@LAPTOP-7H755M1L MINGW64 ~/GitDemo (master)
$ git remote -v
origin https://github.com/Lavanya-Yannam/GitDemo.git (fetch)
origin https://github.com/Lavanya-Yannam/GitDemo.git (push)

LAVANYA@LAPTOP-7H755M1L MINGW64 ~/GitDemo (master)
$ git push origin master
remote: Permission to Lavanya-Yannam/GitDemo.git denied to Lavanya-Yannam.
fatal: unable to access 'https://github.com/Lavanya-Yannam/GitDemo.git/': The requested URL retur
ned error: 403

LAVANYA@LAPTOP-7H755M1L MINGW64 ~/GitDemo (master)
$ git push origin master
info: please complete authentication in your browser...
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 283 bytes | 141.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/Lavanya-Yannam/GitDemo.git
 * [new branch]      master -> master

LAVANYA@LAPTOP-7H755M1L MINGW64 ~/GitDemo (master)
$

```

